Apurva Rai

Ph.D. Candidate, Department of Computer Science, University of California, Irvine
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Research interests: Cryptography — authenticated key exchange, password-authenticated key exchange (PAKE), biometric authentication, deniable authentication, secure messaging, authentication in the short authenticated string (SAS) model.

EDUCATION

University of California, Irvine — Irvine, CA
Ph.D. in Computer Science — September 2021 – present (expected June 2027)

- Dissertation: "Authenticated Key Agreement without Trusted Third Parties: From Short Authenticated Strings to Biometrics"
- Advisor: Prof. Stanislaw Jarecki · Advanced to candidacy March 2025

University of Kansas (Honors Program) — Lawrence, KS
B.S. in Computer Science, magna cum laude — May 2021
B.A. in Mathematics, magna cum laude

HONORS & AWARDS

Dissertation Fellowship, University of California, Irvine	Fall 2026 – Winter 2027
Undergraduate Research Award in Mathematics, University of Kansas	Spring 2021
Engineering Undergraduate Research Fellowship, University of Kansas	Spring 2020 – Spring 2021
Eta Kappa Nu Underclassman Achievement Award in Discrete Structures	Fall 2019
IEA Scholar, University of Kansas — full-tuition award given to 25 international students annually	2017 – 2021

PUBLICATIONS

Publications [C1–C3] follow cryptography's alphabetical authorship convention; I was a lead contributor to each.

Peer-reviewed conference papers

[C1] Gu, Y., Jarecki, S., Nazarian, P., **Rai, A.** "Security without Trusted Third Parties: VRF-Based Authentication with Short Authenticated Strings." *ASIACRYPT 2025*. ePrint 2025/1654 · doi.org/10.1007/978-981-95-5096-8_3

[C2] Jarecki, S., Nazarian, P., **Rai, A.** "X3DH with Deniable Authentication without Trusted Third Parties." *SCN 2026*. ePrint 2026/522

[C3] Jarecki, S., **Rai, A.** "SAS-Model Authentication of Group Key Agreement without Extra Communication Flows." *CANS 2026*, to appear.

Under submission

[S1] Xu, J., **Rai, A.**, Nazarian, P., Jarecki, S., Harris, I. "Asymmetric Fuzzy PAKE for Remote Biometric Authentication with Protection of Server-held Biometric Data." Under submission.

Journal articles (undergraduate research)

[J1] Mahmud, N., MacGillivray, A., **Rai, A.**, Patterson, J., Gharaibeh, A., El-Araby, E., Shaw, H., Cooper, L. "Combining Quantum Key Distribution with Chaotic Systems for Free-Space Optical Communications." *Quantum Information Processing* 20, 354 (2021). doi.org/10.1007/s11128-021-03299-3

[J2] Mahmud, N., Haase-Divine, B., Kuhnke, A., **Rai, A.**, MacGillivray, A., El-Araby, E. "Efficient Computation Techniques and Hardware Architectures for Unitary Transformations in Support of Quantum Algorithm Emulation." *Journal of Signal Processing Systems* 92, 1017–1037 (2020). doi.org/10.1007/s11265-020-01569-4

[J3] Mahmud, N., Haase-Divine, B., MacGillivray, A., Srimoungchanh, B., Kuhnke, A., Blankenau, N., **Rai, A.**, El-Araby, E. "Modifying Quantum Grover's Algorithm for Dynamic Multi-Pattern Search on Reconfigurable Hardware." *Journal of Computational Electronics* 19, 1215–1231 (2020). doi.org/10.1007/s10825-020-01489-3

TALKS

- UC Irvine cryptography seminar: multiple talks presenting my own papers and recent papers from the literature.

RESEARCH EXPERIENCE

Graduate Student Researcher, University of California, Irvine September 2021 – present

Advisor: Prof. Stanislaw Jarecki

- Authentication without trusted third parties: SAS-model message authentication and key exchange for two-party, secure-messaging (X3DH), and group settings [C1–C3]; asymmetric fuzzy PAKE for remote biometric authentication [S1].
- Implemented the full protocol suite for [S1], including optimized precomputed ristretto255 scalar multiplication in C for the paper's performance evaluation.

Undergraduate Researcher, University of Kansas 2019 – 2021

- Quantum-algorithm emulation on reconfigurable hardware, Prof. Esam El-Araby's lab (2019–2021): co-authored [J1–J3]; the group's code ran on a swarm of NASA satellites, and two NASA engineers co-authored [J1].
- GPU-accelerated integer factoring and elliptic-curve computation, Engineering Undergraduate Research Fellow with Prof. Michael Branicky (2020–2021).
- Galois theory and the General Number Field Sieve, Undergraduate Mathematics Research Fellow with Prof. Emily Witt (Spring 2021).

TEACHING EXPERIENCE

Instructor of Record, University of California, Irvine Summer 2026

ICS 6B: Boolean Logic and Discrete Structures

- Taught 20 students over a full 10-week session: designed the course, wrote all exams and assignments, delivered all lectures, and supervised one teaching assistant.

Graduate Teaching Assistant, University of California, Irvine Fall 2022 – Spring 2026

11 quarters: ICS 6D Discrete Mathematics for Computer Science (5), ICS 6B Boolean Logic and Discrete Structures (4), CS 167 Applied Cryptography (1), CS 201P Computer Security, graduate lab (1)

- Head TA for ICS 6D (Winter 2026), leading a team of 8 TAs and helping coordinate discussion content and all exams; co-head TA for four other quarters, guiding the TAs who taught discussion sections while writing the exams.
- Several quarters had over 400 students. Also helped set up the course structure for ICS 6B and ICS 6D, substitute-taught lectures, and led discussion sections for CS 167, teaching cryptographic content directly.

Computer Science Reader, University of California, Irvine Fall 2021 – Spring 2022

Discrete Mathematics, Cryptographic Protocols, Algorithms

Summer Teaching Apprenticeship Program, UCI Division of Teaching Excellence and Innovation Summer 2026

Certificate in evidence-based pedagogy

MENTORING & ADVISING

Co-supervisor, M.S. Capstone Project (4 students), University of California, Irvine December 2025 – June 2026

- Supervised a Master's capstone team implementing the deniable X3DH protocol from [C2]: guided students through the intent of the protocol's algorithms, reviewed and debugged their code for fidelity to the paper, and identified idealizations in the paper that required concrete instantiation.

ACADEMIC SERVICE

- External reviewer (subreviewer): *EUROCRYPT* 2026, *CRYPTO* 2025, *EUROCRYPT* 2025.

SOFTWARE

- Biometric-PAKE: protocol implementation and error-rate experiments for [S1]. github.com/Jinyaox/Biometric-PAKE

SKILLS

- **Programming:** C, Rust, Python (NumPy, Numba), C++, CUDA C, Java, JavaScript, LaTeX.